SWICTHING AND VLANS

1.Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

ANS:

Network Diagram: Devices Connected to a Switch Using VLANs

Managed Network Switch

VLAN 10 and VLAN 20 configured

Smart TV

VLAN 10

Streaming

My Laptop

VLAN 20

Personal

Friend's Phone

VLAN 20

Personal

VLAN Assignment Table

Device	VLAN ID	Purpose
Smart TV	VLAN 10	Streaming
My Laptop	VLAN 20	Personal devices
Friend's Phone	VLAN 20	Personal devices

Explanation

- VLAN 10: Separates the smart TV's streaming traffic from personal devices.
- VLAN 20: Groups the laptop and friend's phone into the same logical network.
- Switch: Uses VLAN settings to separate devices into different logical networks, even when they connect to the same physical switch.

2.Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify

**that PCs in different VLANs cannot ping each other.

Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.**

ANS:

Practical: Configure VLAN 10 and VLAN 20 in Cisco Packet Tracer

Aim: To create two VLANs, assign five PCs to them, and verify that PCs in different VLANs cannot communicate without inter-VLAN routing.

1. Network topology

Cisco Switch (2960)

VLAN 10

192.168.10.0/24

PC0 — .11

PC1 — .12

PC2 — .13

VLAN 20

192.168.20.0/24

PC3 — .11

PC4 — .12

Connect all five PCs to separate switch ports using copper straight-through cables.

2. Assign IP addresses to PCs

PC	VLAN	IP Address	Subnet Mask
PC0	10	192.168.10.11	255.255.255.0
PC1	10	192.168.10.12	255.255.255.0
PC2	10	192.168.10.13	255.255.255.0
PC3	20	192.168.20.11	255.255.255.0
PC4	20	192.168.20.12	255.255.255.0

In Packet Tracer, open each PC, select Desktop → IP Configuration, and enter its IP address and subnet mask. Leave the default gateway blank for this test.

3. Configure the switch

Open the switch CLI and enter these commands:

enable

```
configure terminal

vlan 10
name VLAN10
exit

vlan 20
name VLAN20
exit

interface range fastEthernet 0/1-3
switchport mode access
switchport access vlan 10
exit

interface range fastEthernet 0/4-5
switchport mode access
switchport access vlan 20
exit

end
show vlan brief
```

Explanation:

- vlan 10 and vlan 20 create the two VLANs.
- name assigns a readable name to each VLAN.
- interface range selects multiple switch ports at once.
- switchport mode access configures each port as an access port.
- switchport access vlan assigns the port to the selected VLAN.
- show vlan brief verifies VLAN membership.

4. Test connectivity

On each PC, open Desktop → Command Prompt.

Test PCs within VLAN 10:

ping 192.168.10.12

Expected result: Successful replies when testing from PC0 to PC1.

Test PCs within VLAN 20:

ping 192.168.20.12

Expected result: Successful replies when testing from PC3 to PC4.

Test PCs in different VLANs:

ping 192.168.20.11

Run this from PC0 (192.168.10.11).

Expected result: Ping fails, because the two VLANs are separate networks and no router or Layer 3 switch has been configured to route between them.

5. Verification table

Test	**Expected Result**
PC0 → PC1 (VLAN 10)	Success
PC3 → PC4 (VLAN 20)	Success
PC0 → PC3 (different VLANs)	Failure
show vlan brief	Ports assigned to the correct VLANs

3.In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

ANS:

Practical: Moving a PC from VLAN 10 to VLAN 20

Aim: To change a PC's VLAN membership using Cisco Packet Tracer and test network connectivity before and after the change.

1. Network Setup

- Switch: Cisco 2960
- PC0 connected to switch port Fa0/1
- PC1 connected to switch port Fa0/2
- PC2 connected to switch port Fa0/3

PC	**IP Address**	**Initial VLAN**

PC0 192.168.10.10/24 VLAN 10

PC1 192.168.10.20/24 VLAN 10

PC2 192.168.20.20/24 VLAN 20

2. Before Configuration

Enter these commands in the switch CLI to assign PC0 and PC1 to VLAN 10 and PC2 to VLAN 20.

```
enable
configure terminal

vlan 10
exit
vlan 20
exit

interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit

interface fastEthernet 0/2
switchport mode access
switchport access vlan 10
exit

interface fastEthernet 0/3
switchport mode access
switchport access vlan 20
exit

end
```

show vlan brief

Expected before-change result: Fa0/1 and Fa0/2 belong to VLAN 10, while Fa0/3 belongs to VLAN 20.

3. Test Before the Change

From PC0, open Desktop → Command Prompt and run:

ping 192.168.10.20

Expected result: Successful replies from PC1 because both PCs are in VLAN 10 and the same IP subnet.

4. Move PC0 to VLAN 20

Run these commands on the switch:

enable

configure terminal

interface fastEthernet 0/1

switchport access vlan 20

exit

end

show vlan brief

Explanation: The command switchport access vlan 20 changes the access port Fa0/1 from VLAN 10 to VLAN 20.

5. Update PC0's IP Address

Because VLAN 20 uses a different subnet, change PC0's IP settings in Desktop → IP Configuration:

Setting	Before	After
VLAN	10	20
IP Address	192.168.10.10	192.168.20.10
Subnet Mask	255.255.255.0	255.255.255.0

6. Test After the Change

From PC0, run:

ping 192.168.20.20

Expected result: Successful replies from PC2, after ARP resolution.

Then test PC0 against PC1:

ping 192.168.10.20

Expected result: The ping fails because PC0 is now in VLAN 20 with IP address 192.168.20.10, while PC1 remains in VLAN 10. No inter-VLAN routing is configured.

7. Final Verification Table

Test	**Expected Result**
PC0 → PC1 before moving	Success
PC0's switch port after moving	VLAN 20
PC0 → PC2 after moving	Success
PC0 → PC1 after moving	Failure

**4.Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.**

ANS:

Cisco Packet Tracer – Assign 10 Ports to VLAN 30

Commands:

enable

configure terminal

vlan 30

name VLAN30

exit

interface range fastEthernet 0/1-10

switchport mode access

switchport access vlan 30

exit

end

show vlan brief

Explanation:

- vlan 30 creates VLAN 30.

- interface range fastEthernet 0/1-10 selects ports 1 to 10 together.
- switchport access vlan 30 assigns all 10 ports to VLAN 30.
- show vlan brief verifies the VLAN configuration.

Result: FastEthernet ports 0/1 to 0/10 are assigned to VLAN 30 without repeating commands for each port.